Quantifying Predictive Uncertainty: A Probabilistic Framework for Classifier Evaluation

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***Abstract*—Evaluating the performance of classifiers is crucial in machine learning, particularly in applications where the reliability and interpretability of predictions are paramount for decision-making. Traditional performance measures (e.g., accuracy) rely on binarising probabilistic outputs into hard labels, which overlooks the uncertainty inherent in classifier predictions and can overstate performance—leading to incomplete or misleading assessments. This work develops a probabilistic evaluation framework centred on the probability-mass confusion matrix ($CM^\star$) and on a novel threshold-free decomposition of this matrix into decisive and residual components $CM^\star$ (λ_v and λ_u). These quantities describe how much of the total predictive probability mass is concentrated in the top-1 components and how much remains distributed across non-top classes, providing a complementary diagnostic for model comparison and deployment assessment. Empirically, across 27 UCI datasets, probability-aware scores computed from $CM^\star$ are typically lower than hard-label scores because they account for diffuse predictive mass beyond the top-1 class. Using a Random Forest baseline, the mean accuracy (Acc) is 0.87, while the probabilistic accuracy ($Acc^\star$) and the area under the multi-class classification performance (MCP) curve average 0.77 and 0.71, respectively. Moreover, higher certainty shows a strong monotonic association with smaller nominal–robabilistic performance gaps, a pattern that persists with isotonic-calibrated multinomial logistic regression. These findings indicate that the two frameworks answer complementary questions: CM summarises realised argmax decisions, whereas $CM^\star$ and its decisiveness/residual-dispersion decomposition characterise the concentration and class-pair allocation of the underlying predictive probability mass. Accordingly, $CM^\star$-based measures should complement conventional metrics and be interpreted jointly with diagnostics of probability quality.**

***Index Terms*—Uncertainty, classification, confusion matrix, probabilistic confusion matrix, performance measures.**

I. Introduction

CLASSIFICATION is a fundamental task in machine learning, where the goal is to assign a class label to each instance in a dataset based on input features. A wide range of classification algorithms have been developed and extensively used across various fields, including healthcare, finance, and scientific research. Evaluating the performance of these classifiers is crucial for understanding their effectiveness and reliability in practical applications [1].

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With the rise of more sophisticated models (e.g., deep learning-based ones), explaining closed-box models has become a popular yet often inadequate practice [2], partially diverting attention from the factors that truly explain the performance of predictive models.

Traditionally, classification performance is assessed using confusion matrices and derived measures such as accuracy [3], F-score [4], and the Matthews correlation coefficient (MCC) [5], among others [6]. These measures, however, often rely on hard classification decisions, where each instance is definitively assigned to a single class. This approach overlooks the probabilistic nature of many classification models that provide class membership probabilities rather than categorical labels. As a result, conventional confusion matrices may overestimate the performance of classifiers by ignoring the inherent uncertainty in predictions.

The basic concept for expressing degrees of uncertainty is probability. Classifiers not only produce predictions (i.e., the class assigned to a new, unseen test sample), but also the probabilities associated with each class, from which the predictions are derived (typically by selecting the class with the highest probability). Evaluating these probabilities directly (rather than only the final predictions) provides a richer perspective for addressing the classification problem.

The motivation of this work is illustrated in Fig. 1. Consider a binary dataset with classes A and B and $n = 20$ samples (10 from each class). For each instance e_i, let the classifier output the class-probability vector $\rho(e_i) = (q_{i,A}, q_{i,B})$, with $q_{i,A} + q_{i,B} = 1$, and let $\varepsilon \in (0, 0.5]$ control the concentration of the predictive distribution. For simplicity, we take the same ε for all instances and assume that the classifier assigns slightly more probability mass to the true class:

$$q_{i,y_i} = \frac{1}{2} + \varepsilon, \qquad q_{i,\neg y_i} = \frac{1}{2} - \varepsilon$$

so that, for any $\varepsilon > 0$, the winner class $\arg\max_j q_{i,j}$ coincides with y_i for every sample.

Under this construction, the standard confusion matrix is identical for all $\varepsilon > 0$ because the induced *hard* decision does not change: each predictive distribution $\rho(e_i)$ is converted into a one-hot class assignment by selecting its argmax, and this assignment remains correct for every instance. Hence the resulting hard-label evaluation reports perfect accuracy (i.e., the proportion of correctly classified instances equals 1) for all $\varepsilon > 0$. By contrast, the probabilistic confusion matrix $CM^\star$ (defined in Section III, Definition 6, (2)) varies continuously

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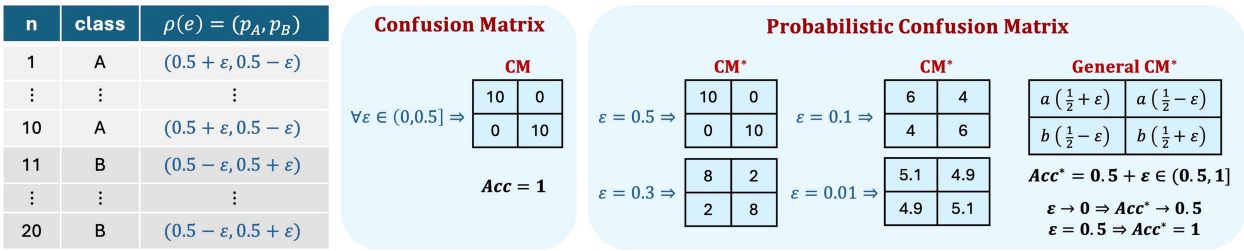


Fig. 1. Standard vs. probabilistic confusion matrices for a binary dataset ({A,B}) with 20 samples (10 per class). The standard confusion matrix (CM) remains unchanged across all values of ε (small variations on probability), always yielding perfect accuracy. In contrast, the probabilistic confusion matrix (CM*) depends on ε: as ε decreases, the accuracy derived from CM* drops from 1 toward 0.5, reflecting increased prediction uncertainty.

with ε, because it aggregates the full probability mass assigned to each class. In particular, the probabilistic accuracy satisfies

$$Acc^\star = \frac{1}{n}\sum_{i=1}^{n} q_{i,y_i} = \frac{1}{2} + \varepsilon$$

so $Acc^\star$ decreases as ε approaches 0 and $\lim_{\varepsilon\to 0} Acc^\star = \frac{1}{2}$. Thus, Fig. 1 shows that hard-label evaluation can be insensitive to declining decisiveness when the argmax decision remains unchanged, whereas CM*-based evaluation captures that decline through the redistribution of probability mass.

The underlying reason is that CM records only binarised decisions, discarding the magnitude of the predictive probabilities before evaluation. While discretisation is often useful for simplifying inputs, applying it to classifier output probabilities removes information that is directly relevant to reliability assessment, since probability vectors are inherently numerical and their concentration (or dispersion) is informative. Consequently, measures computed from CM may overstate reliability relative to their CM*-based counterparts whenever predictions become less decisive without changing the argmax. To place our proposal in context, we briefly recall several established probability-aware evaluation paradigms and clarify how they relate to (and differ from) probability-mass accounting via CM*.

In particular, the literature offers probability-aware evaluation tools that mitigate, in different ways, the information loss induced by hard decisions, most notably ROC [7] and precision–recall (PR) [8] curves (and their area-under-the-curve summaries, AU(·)). These are threshold-sweeping procedures: rather than fixing a single operating point, they vary a decision threshold applied to a scalar score and evaluate the induced trade-off between error types across the resulting operating points. Consequently, when ROC/PR curves are computed by thresholding a scalar score, they are driven primarily by the ranking induced by that score (i.e., by discrimination). In particular, in the absence of ties, any strictly monotone transformation of the score preserves the ranking and therefore leaves the ROC/PR curves unchanged; temperature scaling provides a canonical example of a transformation that can alter probability sharpness while preserving the ordering of scores in the binary case.

In contrast, the Multi-class Classification Performance (MCP) curve [9] is value-sensitive: it is not determined solely by the ordering of a single score, but by a distributional discrepancy (via the Hellinger distance) between the true and the predicted class-probability vectors. As a result, recalibration procedures that modify the numerical probability mass allocation (e.g., by changing sharpness or calibration) will, in general, alter the Hellinger distance and hence AU(MCP), even in situations where the instance ranking exploited by ROC/PR remains essentially unchanged.

Beyond ROC/PR-type discrimination analysis and distance-based summaries such as MCP, two further probability-aware paradigms are widely used. First, proper scoring rules and associated calibration diagnostics (e.g., log loss and the Brier score, complemented by reliability diagrams) assess the quality of probabilistic forecasts directly [10], [11], [12]. These tools are indispensable when probabilities are to be interpreted quantitatively; however, they provide global scalar assessments and do not preserve the confusion-matrix structure, offering limited cell-wise interpretability in terms of true–predicted class interactions. Second, reject-option ("doubt") methods introduce abstention through an explicit acceptance threshold on a confidence measure (typically the maximum posterior probability), thereby separating accepted decisions from inconclusive cases [13], [14]; this adds an additional operating parameter that must be selected.

Confusion-like matrices with non-binary entries have also been considered in several settings. Soft-classification validation extends confusion-matrix measures to partial predicted and, in some cases, reference memberships [15]; Wang et al. [16] aggregate predicted class probabilities conditional on the true class, yielding an unnormalised form equivalent to CM*; and other probabilistic variants have been proposed for performance analysis [17].

Taken together, these probability-aware tools address complementary aspects of performance: discrimination across thresholds (ROC/PR), distributional closeness (MCP), probability quality (proper scores/calibration), and abstention under an explicit confidence threshold (reject-option methods). This focus is conceptually different from our objective here: CM* retains the full probability mass assigned to each class at the model's probabilistic operating point, enabling probability-aware counterparts of confusion-matrix-based metrics and an explicit quantification of predictive decisiveness (via λ_v and λ_u), which is not directly provided by threshold-sweeping summaries or single-number proper scores.

The aim of this paper is to develop a new probabilistic framework for classifier evaluation, in which class-probability mass complements hard-label assessment, while making it possible to quantify prediction decisiveness and residual probability dispersion explicitly.

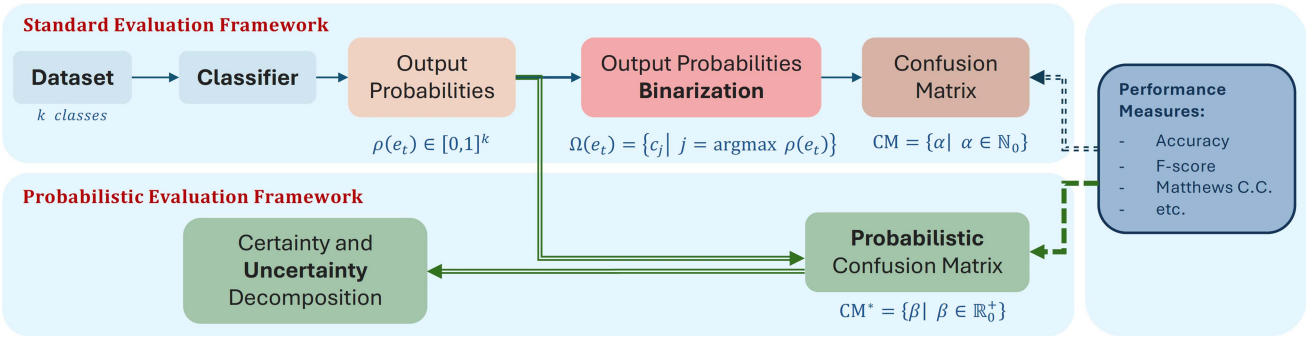


Fig. 2. Standard vs. probabilistic evaluation frameworks.

In summary, this work addresses two main aspects:

1) A unified probability-mass confusion framework, CM*, explicitly positioned relative to prior soft and probabilistic confusion matrices. The framework preserves class-pair structure and supports probabilistic counterparts of confusion-matrix-derived measures.
2) A threshold-free matrix-valued decomposition of CM*, which separates decisive top-1 probability mass from residual dispersion while retaining its class-pair allocation; λ_v and λ_u provide global summaries of these components and connect overall decisiveness with cell-wise class-pair allocations.

Fig. 2 visually summarises the goal of this research. It schematizes the standard evaluation framework, where output probabilities are binarised to build the confusion matrix, and classification performance measures are then applied to this matrix. It also illustrates the probabilistic evaluation framework, in which the probabilistic confusion matrix is obtained directly from the output probabilities, and subsequently decomposed into certainty and uncertainty matrices, with these terms referring specifically to prediction decisiveness and residual probability dispersion, respectively.

The decomposition of the probabilistic confusion matrix into two matrices is a novel perspective designed to quantify the extent to which a classifier's performance is driven by confident predictions versus uncertain ones. It reveals the proportion of accurate predictions for which the model is genuinely confident. This measure is particularly valuable in high–stakes fields, where the cost of uncertain or incorrect predictions can be substantial. Practitioners may use this framework to complement conventional performance evaluation by examining both hard decisions and the allocation of predictive probability mass. This additional information can support model comparison and selection in applications where the decisiveness of predictions may affect the consequences of subsequent decisions.

The remainder of this paper is organised as follows: Sections II and III provide formal definitions of the confusion matrix and the probabilistic confusion matrix. Section IV introduces the decomposition of probabilistic predictions into certainty and uncertainty components. The concept of probabilistic measure is introduced in Section V. Section VI describes the experimental setup, datasets, and an analysis of results. Finally, Section VII discusses the implications of using probabilistic measures and concludes with potential directions for future research.

II. Confusion Matrix

The confusion matrix is a fundamental concept in machine learning that represents the performance of a classification algorithm in a square matrix format, organised along two dimensions: actual and predicted classes. Each row corresponds to instances of a specific actual class, while each column represents instances assigned to a predicted class. This structure facilitates a detailed analysis of an algorithm's performance, capturing both correct predictions and different types of errors.

The term *confusion matrix*, introduced in 1955, is a table of natural–number counts arising from a forced–choice listening experiment in which listeners guessed 16 English consonants degraded by noise or filtering, with every misidentification tallied [18]. No probability estimates were elicited from participants—unlike modern classifiers, which output real–valued probabilities in [0,1] for each possible class label—yet the confusion matrix remains widely used in its original natural–number form. However, the idea predates 1955: Pearson had already introduced in 1904 the *contingency table* [19], a count–based cross–classification of categorical variables that anticipates the confusion matrix.

Most evaluation metrics used in practice are derived from the hard-label confusion matrix—including Cohen's κ [20], MCC [5], F_β–score [4], the multi–class adaptation of the MCC (R_k) [21], the harmonic mean of per-class sensitivity [22]. These measures rely on binarisation, which compresses rich predictive probability distributions into discrete outcomes, discarding probability mass information and potentially masking uncertainty. In contrast, other performance measures are based on class membership probabilities rather than the confusion matrix. For instance, in binary classification, the area under the Receiver Operating Characteristic (ROC) curve [7], [23], [24], [25], [26], [27] is a widely used metric, while for multi–class problems, the area under the Multi–class Classification Performance (MCP) curve [9], [28] has been proposed. In this context, incorporating predicted probabilities directly into a confusion-matrix structure remains comparatively underexplored, despite providing a natural and interpretable way to capture decisiveness and dispersion when assessing reliability.

We consider a multi-class classification problem with label set $\mathbb{L} = \{c_1, \ldots, c_k\}$ (k classes) and a dataset D of n examples $e_i = (x_i, y_i)$, where $x_i \in \mathbb{R}^m$ (m variables) and $y_i \in \mathbb{L}$. A classifier is a function $\Omega : \mathbb{R}^m \to \mathbb{L}$, learned from D, whose goal is to predict the label of a previously unseen example; in practice, Ω is induced by a predictive model ρ that outputs either a class label or, in the probabilistic setting considered here, a class-probability vector over $\mathbb{L}$.

Definition 1 (Dataset): A dataset is a finite set

$$D = \{(x_i, y_i)\}_{i=1}^{n} \subset \mathbb{R}^m \times L.$$

When arranged in tabular form, its feature matrix has n rows and m columns, together with one additional label column.

Example 1 (Dataset): Consider a dataset D with six instances, m variables, and three labels {A, B, C}.

$$D_{6,m+1} = \begin{bmatrix} \cdots & A \\ \cdots & A \\ \cdots & A \\ \cdots & B \\ \cdots & B \\ \cdots & C \end{bmatrix}$$

Definition 2 (Ground–truth Matrix): Let $T_{n,k}$ be the *Ground–truth Matrix* that contains the class label of each example from the dataset D. $T_{n,k}$ can be defined as a matrix of n rows (instances) and k columns (classes), where each row represents the class label expressed as a one–hot vector.

$$T_{n,k} = \begin{bmatrix} t_{1,1} & \ldots & t_{1,k} \\ \vdots & & \vdots \\ t_{n,1} & \ldots & t_{n,k} \end{bmatrix}$$

where, for each instance $e_i = (x_i, y_i)$,

$$t_{i,j} = \begin{cases} 1, & \text{if } y_i = c_j \\ 0, & \text{otherwise} \end{cases}$$

Example 2 (Ground–truth Matrix): If a dataset has three labels {A, B, C}, and an example e_i belongs to class A, then $t_i = (1, 0, 0)$, i.e., the class A is active (1) and the others are inactive.

The label column {A, A, A, B, B, C} could be transformed into $T_{6,3}$ by means of label binarisation, such as:

$$T_{6,3} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

where each instance is represented in rows, and each class in columns, respectively.

Definition 3 (Prediction Matrix): Let $P_{n,k}$ be the *prediction matrix*, defined by elements $p_{i,j}$ as follows:

$$P_{n,k} = \begin{bmatrix} p_{1,1} & \ldots & p_{1,k} \\ \vdots & & \vdots \\ p_{n,1} & \ldots & p_{n,k} \end{bmatrix}$$

where, for each instance $e_i = (x_i, y_i)$, $p_{i,j}$ is defined as:

$$p_{i,j} = \begin{cases} 1, & \text{if } \Omega_\rho(x_i) = c_j \\ 0, & \text{otherwise} \end{cases}$$

Example 3 (Prediction Matrix): A possible example of P, given the outputs of a model ρ, for the dataset D described earlier, would be:

$$P_{6,3} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

where each row refers to an example, i.e., the model ρ classified the first 4 examples as class A, the next 2 examples as class B, and 0 examples as class C.

Definition 4 (Confusion Matrix): Let $\text{CM}_{k,k}$ be the *Confusion Matrix*, which can be defined as:

$$\text{CM}_{k,k} = T_{n,k}^{\top} P_{n,k} \tag{1}$$

T and P are binary matrices, i.e., all the values $t_{i,j} \in T$ and $p_{i,j} \in P$ are in {0,1}. The confusion matrix CM can be expressed as the product of the transpose of the ground–truth matrix T and the prediction matrix P, where $\forall \alpha_{i,j} \in \text{CM}$, $\alpha_{i,j} \in \mathbb{N}_0$.

Example 4 (Confusion Matrix): In the example, multiplying the transpose $T_{6,3}^{\top}$, and $P_{6,3}$, yields $\text{CM}_{3,3}$.

$$\text{CM}_{3,3} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

III. Probabilistic Confusion Matrix

While the confusion matrix has been a cornerstone for evaluating classification performance, its traditional structure primarily focuses on hard classification decisions, where each instance is definitively assigned to a class. This approach neglects the probabilistic nature of many modern classification models, which often provide a degree of uncertainty associated with their predictions. Ignoring this uncertainty can lead to overestimated performance metrics and an incomplete understanding of model behaviour, particularly in applications where reliable predictions are critical.

For an input vector x_i, the predictive model ρ outputs a class-probability vector $\rho(x_i) = (q_{i,1}, \ldots, q_{i,k})$, with $q_{i,j} \geq 0$ and $\sum_{j=1}^{k} q_{i,j} = 1$. Then, the function Ω is defined as

$$\Omega_\rho(x_i) = c_{j^\star}, \qquad j^\star = \arg \max_{1 \leq j \leq k} q_{i,j},$$

where ρ could be any classification technique (e.g., decision trees, nearest neighbours, neural networks, etc.). The prediction matrix $P_{n,k}$ is then built by binarising the outputs of ρ. However, this binarisation is not required: the prediction matrix can be populated directly with the probability vectors $\rho(x_i)$, thereby retaining the full predictive distribution rather than collapsing it to a one-hot decision.

Hand [29] mentioned the possibility of generalising the confusion matrix, containing probabilities in each of the cells, and defining an estimate of it in terms of proportions. Yacouby and Axman [30] proposed a probabilistic extension of precision, recall and F_1–score in the context of Natural Language Processing (NLP). Under this perspective, we can define a new prediction matrix, $Q_{n,k}$, whose values are directly obtained from the classifier ρ. These studies and the related soft/probabilistic confusion-matrix literature show that using non-binary class memberships in confusion-like structures has precedents. Building on this line of work, we use the complete predictive matrix Q as a class-pair probability-mass representation and subsequently introduce the threshold-free V/U decomposition, which is the central methodological contribution developed in this paper.

Definition 5 (Probabilistic Prediction Matrix): The matrix $Q_{n,k}$ is obtained directly from the classifier ρ, in such a way that each row represents the output probabilities of ρ for each instance e_i, i.e., $q_i = \rho(e_i)$.

$$Q_{n,k} = \begin{bmatrix} q_{1,1} & \cdots & q_{1,k} \\ \vdots & & \vdots \\ q_{n,1} & \cdots & q_{n,k} \end{bmatrix}$$

Most classification performance measures are based on the confusion matrix introduced in Definition 4, which requires the prediction matrix $P_{n,k}$. However, $P_{n,k}$ is typically derived from probabilities, specifically from the probabilistic prediction matrix $Q_{n,k}$ introduced in Definition 5. The process of transforming $Q_{n,k}$ into $P_{n,k}$ is straightforward: each row vector (representing the probabilities of belonging to each class) is converted into a one–hot vector, where 1 is assigned to the class with the highest probability, and 0 to all others. Due to this deterministic transformation, the confusion matrix $\mathrm{CM}_{k,k}$ was aptly referred to as the *forced classification matrix* by Habbema et al. [14].

Example 5 (Probabilistic Prediction Matrix): A possible example of $Q_{6,3}$, given the outputs of ρ for the dataset D, would be:

$$Q_{6,3} = \begin{bmatrix} 0.9 & 0.1 & 0 \\ 0.8 & 0 & 0.2 \\ 0.6 & 0.1 & 0.3 \\ 0.4 & 0.3 & 0.3 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.9 & 0.1 \end{bmatrix}$$

In principle, since $P_{6,3}$ was originally derived from $Q_{6,3}$, the information contained in Q is richer than that in P. The potential of using Q instead of P will be analysed next.

Definition 6 (Probabilistic Confusion Matrix): Let $\mathrm{CM}^\star_{k,k}$ be the *probabilistic confusion matrix*, which can be defined as:

$$\mathrm{CM}^\star{}_{k,k} = T^\top_{n,k} Q_{n,k} \tag{2}$$

where $\forall\ \beta_{i,j} \in \mathrm{CM}^\star{}_{k,k}$, $\beta_{i,j} \in \mathbb{R}^+_0$, i.e., all the values in $\mathrm{CM}^\star$ are non-negative real numbers.

Henceforth, we use the superscript * to denote the probabilistic counterpart of any object—measures, matrices, and other quantities (e.g., $\mathrm{CM}^\star$).

Example 6 (Probabilistic Confusion Matrix): From the previous example, the probabilistic confusion matrix $\mathrm{CM}^\star_{3,3}$ can be obtained by multiplying the transpose $T^\top_{6,3}$ and $Q_{6,3}$.

$$\mathrm{CM}^\star{}_{3,3} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.9 & 0.1 & 0 \\ 0.8 & 0 & 0.2 \\ 0.6 & 0.1 & 0.3 \\ 0.4 & 0.3 & 0.3 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.9 & 0.1 \end{bmatrix}$$

$$= \begin{bmatrix} 2.3 & 0.2 & 0.5 \\ 0.5 & 1.1 & 0.4 \\ 0 & 0.9 & 0.1 \end{bmatrix}$$

$\mathrm{CM}^\star_{3,3}$ differs slightly from $\mathrm{CM}_{3,3}$. While the rows of both matrices sum to the number of examples for each class {A = 3, B = 2, C = 1}, the column sums, which represent the amount of predictions for each class, differ. In CM, the column sums are hard counts derived from definitive class assignments; in $\mathrm{CM}^\star$, they are aggregated probability masses for each class. For class A, the column sum in CM is 4, whereas in $\mathrm{CM}^\star$ it is 2.8. For class B, the column sum in CM is 2, compared to 2.2in $\mathrm{CM}^\star$, and for class C, the column sum in CM is 0, while in $\mathrm{CM}^\star$ it is 1. Notably, the probabilistic distribution of predictions (2.8, 2.2, 1) aligns more closely with the actual class distribution (3, 2, 1) than the hard prediction distribution (4, 2, 0).

The standard confusion matrix $\mathrm{CM} = T^\top P$ depends only on hard decisions, whereas the probabilistic confusion matrix $\mathrm{CM}^\star = T^\top Q$ aggregates the full class-probability vectors in Q. The map $Q \mapsto P$ is therefore many-to-one: distinct probability matrices Q may induce the same hard predictions P (and hence the same CM) while yielding different $\mathrm{CM}^\star$ (as illustrated in Fig. 1). If the predicted probabilities in Q are biased or miscalibrated (e.g., systematically overconfident), this will be reflected in $\mathrm{CM}^\star$-based metrics, whereas it may be invisible after the hard projection $Q \mapsto P$ used to form $\mathrm{CM} = T^\top P$. Accordingly, $\mathrm{CM}^\star$ is more informative whenever predictions are non-degenerate, i.e., whenever at least some instances allocate non-zero probability mass to more than one class.

The two matrices coincide exactly when the predictive distributions are deterministic in the sense that $Q = P$ row-wise (up to the chosen tie convention), in which case $\mathrm{CM}^\star = \mathrm{CM}$. More generally, $\mathrm{CM}^\star$ will be numerically close to CM when most

probability vectors are strongly concentrated on their argmax class (equivalently, when the residual mass away from the maximum is small for most instances). In this regime, hard-label evaluation captures nearly all probability mass and the differences between CM- and CM*-based measures are correspondingly small. Conversely, when predictions are less decisive, CM* departs from CM by allocating non-negligible probability mass to off-diagonal entries even if the argmax decisions (and thus CM) remain unchanged.

The column sums of CM* have a direct interpretation: $\sum_{i=1}^{k} \mathrm{CM}^\star_{i,j} = \sum_{i=1}^{n} q_{i,j}$, namely, the expected number of instances assigned to class j under stochastic sampling from Q. Comparing these predicted marginal frequencies with the empirical class distribution can therefore serve as a descriptive check for systematic prediction skew (over-/under-allocation of probability mass to particular classes). This check is informative about marginal behaviour, but it should not be interpreted as a general calibration assessment: calibration is a conditional property (agreement between predicted probabilities and empirical frequencies at a given probability level).

In the same spirit, discrepancies in precision (e.g., inflated values in CM due to hard assignments) are more naturally interpreted as reflecting differences in decisiveness (probability concentration) between hard-label and probability-mass accounting, and may, in some cases, be consistent with overconfident probability estimates; however, such effects should not be treated as stand-alone evidence of poor calibration [31]. A direct assessment of calibration typically relies on proper scoring rules and reliability diagrams [32], [33], [34], which require binning prediction probabilities.

IV. Certainty and Uncertainty Matrices

Identifying uncertainty from the confusion matrix is crucial for a comprehensive evaluation of classification models. Quantifying uncertainty provides valuable insights into the areas where a model struggles, enables differentiation between confident errors and uncertain predictions, and helps evaluate the potential risks involved in deploying the model [34], [35].

The probabilistic confusion matrix CM* provides information about the classification probabilities, reflecting the accuracy of predictions made by Ω for each test example based on the probabilities generated by the classification model ρ. However, not all probabilities should be treated equally because some correspond to decisive top-1 probability mass, whereas others correspond to residual probability mass distributed across non-top classes. Therefore, decomposing CM* into two matrices –one for certainty and one for uncertainty– can help analyse the degree of certainty and uncertainty in predictions, offering deeper insights into the impact of uncertainty on classification performance.

Within a clinical classification context, Habbema et al. [14] observed that all patients were always allocated to a disease, even when the probability diagnosis expressed significant uncertainty —for instance, when the two largest probabilities were nearly equal. To address this issue, they proposed distinguishing patients with confident diagnoses from those with inconclusive diagnoses by specifying a threshold value for probabilities. Specifically, if the probability exceeded the threshold δ, the patient was allocated to a disease; otherwise, the patient contributed to the *doubt* category in the *classification matrix with doubt*. In this $(k+1)\times k$ matrix, any non-doubt assignment—correct or incorrect—satisfies the acceptance criterion $\max_j q_{rj} \geq \delta$; hence misclassifications correspond to errors made with at least δ posterior support. They also introduced the concept of an *exclusion matrix*, which contained diseases (classes) with probabilities so low (below a specified threshold ϵ) that they could be excluded from further consideration.

These structurally similar matrices contribute to filtering the probabilistic confusion matrix CM*, conserving probabilities larger than δ and removing probabilities smaller than ϵ. However, they do not decompose the probabilistic confusion matrix CM* into two matrices to separately identify certainty and uncertainty. As it will be explained later, the proposed decomposition does not depend on any threshold.

Definition 7 (Decomposition of Q): The matrix $Q_{n,k}$ can be decomposed into two matrices, such that $Q_{n,k} = Q^+_{n,k} + Q^-_{n,k}$, and, for all $q^+ \in Q^+$ and $q^- \in Q^-$:

$$q^+_{i,j} = \begin{cases} q_{i,j}, & \text{if } q_{i,j} = \max_j q_{i,j} \\ 0, & \text{otherwise} \end{cases}$$

$$q^-_{i,j} = \begin{cases} q_{i,j}, & \text{if } q_{i,j} \neq \max_j q_{i,j} \\ 0, & \text{otherwise} \end{cases}$$

The goal of this decomposition is to separate $Q_{n,k}$ into two matrices, $Q^+_{n,k}$ and $Q^-_{n,k}$, where $Q^+_{n,k}$ contains the probability values that lead to predictions (correct or incorrect), and $Q^-_{n,k}$ contains the probability values that do not lead to predictions.

Example 7 (Decomposition of Q):

$$\underbrace{\begin{bmatrix} 0.9 & 0.1 & 0 \\ 0.8 & 0 & 0.2 \\ 0.6 & 0.1 & 0.3 \\ 0.4 & 0.3 & 0.3 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.9 & 0.1 \end{bmatrix}}_{Q_{6,3}} = \underbrace{\begin{bmatrix} 0.9 & 0 & 0 \\ 0.8 & 0 & 0 \\ 0.6 & 0 & 0 \\ 0.4 & 0 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0.9 & 0 \end{bmatrix}}_{Q^+_{6,3}} + \underbrace{\begin{bmatrix} 0 & 0.1 & 0 \\ 0 & 0 & 0.2 \\ 0 & 0.1 & 0.3 \\ 0 & 0.3 & 0.3 \\ 0.1 & 0 & 0.1 \\ 0 & 0 & 0.1 \end{bmatrix}}_{Q^-_{6,3}}$$

The matrix $Q^+_{n,k}$ contains the highest probability values for each row, representing the probabilities that were decisive in producing the matrix P. Thus, these values (0.9, first row; 0.8, second row; ... ; 0.9, sixth row) are 1in $P_{6,3}$ (Example 3). In contrast, the matrix $Q^-_{n,k}$ contains the remaining values that did not correspond to the highest probability in each row, reflecting the uncertainty in the predictions.

For each instance e_i, let $q_{i\cdot}$ denote the predictive class-probability vector and define $\pi_i = \max_j q_{i,j}$. Assume that the max is unique in each row; equivalently, ties are resolved according to a fixed convention. We split $q_{i\cdot}$ into a *decisive* component and a *residual* component by letting Q^+ retain, row-wise, the maximal probability mass and setting $Q^- = Q - Q^+$.

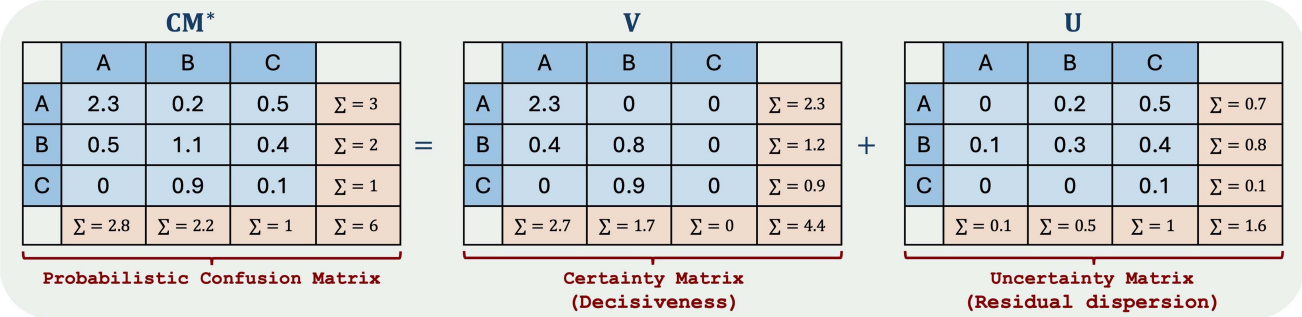


Fig. 3. Probabilistic confusion matrix: decomposition.

Consequently, for each i,

$$\sum_j q^+_{i,j} = \pi_i \qquad \sum_j q^-_{i,j} = 1 - \pi_i$$

In this work, we interpret Q^+ as capturing *predictive certainty* in the sense of probability concentration on the most plausible class, whereas Q^- captures the *dispersion* of the predictive distribution across non-top classes, which we refer to as uncertainty in the sense of *lack of decisiveness*. In particular, if the predictive distribution is one-hot (i.e., $\pi_i = 1$), then Q^- vanishes and uncertainty is null; conversely, as probabilities become more diffuse, $1-\pi_i$ increases and so does the residual mass.

This notion should be distinguished from conventional uncertainty quantification (UQ), where the aim is to attribute uncertainty to aleatoric versus epistemic sources [36], as well as from broader UQ (including non-probabilistic convex modelling [37], [38], [39]) whose goal is to characterise uncertainty sets or reliability under limited information. Here we do not seek to identify the origin of uncertainty; rather, we quantify the certainty of the prediction through the concentration of the predictive categorical distribution.

It is important to emphasize that in the confusion matrix CM, only the values from $Q^+_{n,k}$ are considered (transforming non–zero values into 1), while the values from $Q^-_{n,k}$ are excluded, i.e., this valuable information is lost. Consequently, the confusion matrix focuses solely on the probabilities that lead to the most confident predictions (not necessarily accurate), quantitatively disregarding those associated with lower certainty. This selective consideration highlights the model performance based on its most confident decisions, while the excluded values could provide meaningful insights into areas of uncertainty.

Definition 8 (Decomposition of CM*): Let $\mathrm{CM}^\star$ be the probabilistic confusion matrix. $\mathrm{CM}^\star$ can be decomposed into two matrices $\mathrm{CM}^\star = \mathrm{V} + \mathrm{U}$, where V and U are computed as follows:

$$\mathrm{CM}^\star_{k,k} = T^\top_{n,k}\left(Q^+_{n,k} + Q^-_{n,k}\right) = \mathrm{V}_{k,k} + \mathrm{U}_{k,k} \tag{3}$$

Definition 9 (Certainty Matrix (V)): The matrix $\mathrm{V}_{k,k}$ collects the contribution to $\mathrm{CM}^\star$ of the decisive probability mass (the argmax component) made by the classifier ρ. It represents the predictive certainty of the classifier, as it assigns a class to a test example with higher probability than to other classes.

Definition 10 (Uncertainty Matrix (U)): The matrix $\mathrm{U}_{k,k}$ collects the contribution to $\mathrm{CM}^\star$ of the residual probability mass (dispersion across alternatives) made by the classifier ρ. It represents the predictive uncertainty of the classifier, as the probabilities present in this matrix correspond to classes different from the class predicted by ρ.

Example 8 (Decomposition of CM*): The $\mathrm{CM}^\star_{3,3}$ obtained in Example 6 can be decomposed into the certainty $\mathrm{V}_{3,3}$ and uncertainty $\mathrm{U}_{3,3}$ matrices, as shown in Fig. 3, with row and column totals denoted by Σ.

Example 9 (Sensitivity. Redistribution of mass between V and U): Consider $k = 3$ classes $\{A, B, C\}$ and a single instance e with true label $y = A$ (so $T = (1, 0, 0)$). Let the predicted probabilities be

$$q = (q_A, q_B, q_C) = (0.60,\ 0.25,\ 0.15).$$

Using the row-wise decomposition $Q = Q^+ + Q^-$, we keep the maximal (top-1) mass in Q^+ and set $Q^- = Q - Q^+$, hence

$$q^+ = (0.60,\ 0,\ 0), \qquad q^- = (0,\ 0.25,\ 0.15).$$

Therefore, only the row of the true class is populated and

$$V_{A\cdot} = q^+, \qquad U_{A\cdot} = q^-, \qquad \mathrm{CM}^\star_{A\cdot} = V_{A\cdot} + U_{A\cdot} = q.$$

(i) Transfer between V and U (change in decisiveness). For a small $\delta > 0$ that does not change the argmax,

$$(0.60, 0.25, 0.15) \mapsto (0.60 + \delta,\ 0.25 - \delta,\ 0.15),$$

so V_{AA} increases by δ and U_{AB} decreases by δ (all other entries unchanged): exactly δ mass moves from U to V.

(ii) Redistribution within U (change in the pattern of uncertainty). Keeping the top-1 probability fixed but redistributing the residual mass, e.g.

$$(0.60, 0.25, 0.15) \mapsto (0.60,\ 0.20,\ 0.20),$$

leaves V unchanged, while U_{AB} decreases by 0.05 and U_{AC} increases by 0.05: the total uncertainty mass $1 - \max_j q_j$ is preserved, but reallocated across off-diagonal cells.

In summary, V responds only to changes in top-1 mass (decisiveness), whereas U captures the complementary residual dispersion and its allocation across alternative classes.

In Fig. 4, the CM calculated in Example 4 is depicted for clarity, with row and column totals also denoted by Σ. To illustrate the impact of uncertainty on performance assessment, we report

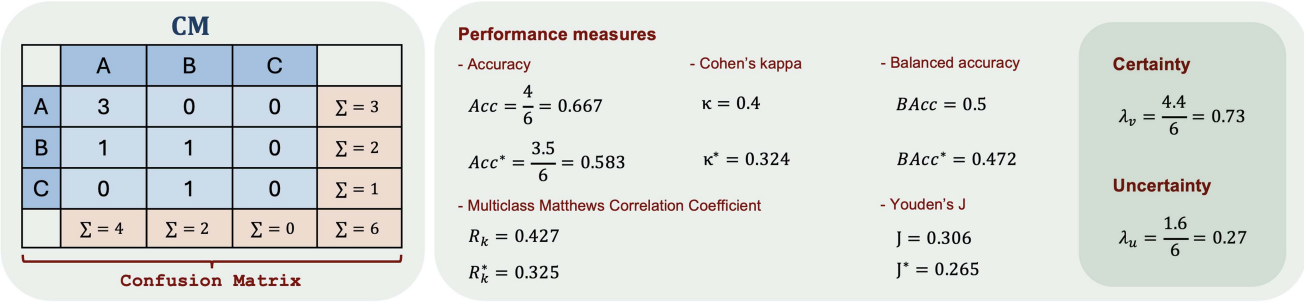


Fig. 4. Performance measures for standard (CM) and probabilistic (CM*) confusion matrices.

several standard multi–class measures for both CM and CM*: accuracy, Cohen's κ, balanced accuracy, the multi–class MCC and Youden's J. In general, the measures computed from the standard CM are optimistic (i.e., they usually overestimate performance) relative to their probabilistic counterparts computed from CM*.

The computation induced by $\mathrm{CM}^\star = T^\top Q$ and by the V/U decomposition admits transparent complexity bounds. Once the probability matrix $Q \in \mathbb{R}$ is available, forming CM* has time complexity $O(nk)$, since each of the n instances contributes a length-k probability vector that is accumulated into the row indexed by its true class. The row-wise split $Q = Q^+ + Q^-$ and the subsequent constructions $\mathrm{V} = T^\top Q^+$ and $\mathrm{U} = T^\top Q^-$ also have time complexity $O(nk)$. By contrast, the standard confusion matrix $\mathrm{CM} = T^\top P$ has time complexity $O(n)$, because each instance contributes a single increment to one cell via its hard prediction.

V. Probabilistic Measures

There are numerous measures to assess the classification performance. To address the general multi-class setting, we will focus on measures that can inform about the overall performance of classifiers, which can be directly obtained from the confusion matrix $\mathrm{CM}_{k,k}$.

Definition 11 (Classification performance measure): A classification performance measure is a function ψ defined as

$$\psi : (\mathbb{N}_0)^{k\times k} \longrightarrow \mathbb{R} \quad such that \psi(\mathrm{CM}_{k,k}) = r \in \mathbb{R} \tag{4}$$

Definition 12 (Probabilistic classification performance measure): A probabilistic classification performance measure is a function ϕ defined as

$$\phi : (\mathbb{R}_0^+)^{k\times k} \longrightarrow \mathbb{R} \quad such that \phi(\mathrm{CM}^\star_{k,k}) = s \in \mathbb{R} \tag{5}$$

Any confusion-matrix-based measure ψ that is well defined on non-negative real-valued entries can be extended as a probabilistic classification performance measure ϕ by replacing the confusion matrix $\mathrm{CM}_{k,k}$ with the probabilistic confusion matrix $\mathrm{CM}^\star_{k,k}$.

Accuracy (or the Finley's index [3]) is by far the most popular measure of classification performance for multi–class datasets. It quantifies the proportion of samples correctly classified by the system. This measure is appealing due to its simplicity but does not account for the severity or magnitude of errors [40]. Accuracy is a particular instance of ψ and is defined by:

$$Acc = \frac{\sum_{i=1}^{k} \alpha_{i,i}}{\sum_{i=1}^{k}\sum_{j=1}^{k} \alpha_{i,j}} \tag{6}$$

where k is the number of classes (number of rows or columns of the matrix), and $\alpha_{i,j} \in \mathrm{CM}_{k,k}$, for all $i, j \in \{1, \ldots, k\}$.

While accuracy measures the overall correctness of the classifications, it does not account for the quality of the predicted probabilities. For example, probabilities close to 1 and probabilities slightly above $\frac{1}{k}$ might both yield correct hard decisions, but differ significantly in their quality [29]. This aspect, highlighted in Fig. 1, is not considered when calculating the accuracy from the CM.

Example 10 (Accuracy): From the $\mathrm{CM}_{3,3}$ in Example 4, the accuracy is $Acc = \frac{4}{6} = 0.667$, as shown in Fig. 4.

Example 11 (Probabilistic accuracy): From the $\mathrm{CM}^\star_{3,3}$ in Example 6, the probabilistic accuracy is $Acc^\star = \frac{3.5}{6} = 0.583$, as shown in Fig. 4.

The contrast between Acc and $Acc^\star$ can be interpreted as a quantitative measure of how much the evaluation is affected by binarising probabilities. The gap $Acc - Acc^\star$ is small when predictions are highly decisive (most mass concentrates on the true class), and it grows when predictions are correct but weakly supported (probability mass is spread across competing classes). This provides an intuitive bridge between decision-level correctness and probabilistic support.

When computing performance measures for the CM* it is useful to quantify the contribution of certainty and uncertainty, as expressed in matrices V and U. These contributions can be expressed as probability-mass weights.

Let the matrix $\mathrm{CM}^\star_{k,k} = \mathrm{V}_{k,k} + \mathrm{U}_{k,k}$, where $\beta_{i,j} = v_{i,j} + u_{i,j}$, with $\beta_{i,j} \in \mathrm{CM}^\star_{k,k}$, $v_{i,j} \in \mathrm{V}_{k,k}$, and $u_{i,j} \in \mathrm{U}_{k,k}$, for all $i, j \in \{1, \ldots, k\}$. The certainty and uncertainty weights are defined next.

Definition 13 (Certainty/decisiveness weight (λ_v)): The certainty weight is defined as

$$\lambda_v = \frac{\sum_{i,j} v_{i,j}}{\sum_{i,j} \beta_{i,j}} \tag{7}$$

Definition 14 (Uncertainty/dispersion weight (λ_u)): The uncertainty weight is defined as

$$\lambda_u = \frac{\sum_{i,j} u_{i,j}}{\sum_{i,j} \beta_{i,j}} \tag{8}$$

Building on the row-wise decomposition $Q = Q^+ + Q^-$, where Q^+ retains the maximal probability mass and Q^- captures the residual dispersion, the weights admit the closed-form expressions

$$\lambda_v = \frac{1}{n}\sum_{i=1}^{n} \max_j q_{i,j} \qquad \lambda_u = 1 - \lambda_v$$

Thus, λ_v represents the mean top-1 probability (decisiveness), whereas λ_u represents the complementary mean residual mass (dispersion). Importantly, these quantities are outcome-agnostic: they depend only on the predictive distributions and not on the true labels. They are therefore not calibration metrics nor error decompositions in the sense of Brier-score reliability–resolution–uncertainty analyses. Their role is diagnostic rather than evaluative: they quantify how concentrated the predictive distributions are, independently of whether those probabilities are statistically well calibrated.

In the limiting case of perfectly decisive predictions (one-hot distributions), $\lambda_u = 0$ and all probability mass lies in V. Conversely, as predictive distributions become more diffuse, λ_u increases and a larger fraction of CM* is allocated to U.

The contribution of certainty and uncertainty to any probabilistic performance measure $\phi(\text{CM}^\star)$ is therefore modulated by λ_v and λ_u. When U carries little mass (equivalently, λ_u is small), metrics computed on CM* tend to be close to those computed on V for trace-like measures (e.g., accuracy).

The weights also admit an immediate probabilistic interpretation. For example, if $\lambda_v = 0.73$ (as in Fig. 4), then 73% of the total probability mass lies in the certainty matrix V, while the remaining 27% lies in the residual matrix U. This reflects decisiveness rather than accuracy: high certainty does not necessarily imply correct predictions. The weights are sensitive to both the distribution of probabilities across classes and their absolute values: a prediction assigning probability 0.9 to the true class contributes strongly to V, whereas a prediction with top probability 0.4 and similar competitors increases U, signalling limited decisiveness—a scenario that becomes more common as the number of classes grows and class overlap increases.

VI. Experimental Analysis

The experimental analysis is designed to evaluate the benefits of using the probabilistic confusion matrix CM* compared to the standard confusion matrix CM, and to analyse the contribution of the certainty–uncertainty decomposition to model explainability. Both the choice of classifier and the performance metrics are purely illustrative. The framework is model- and metric-agnostic; the experiments use representative classifiers and probability-aware measures to illustrate its behaviour.

The study involves 27 datasets from the UCI Machine Learning Repository [41], varying in the number of samples (100 to 19,020), number of variables (4 to 166), and number of classes (2 to 30) (see Table I). All experiments use stratified 10-fold cross-validation. Two multi–class classification performance measures were selected: accuracy (Acc, derived from CM, and probabilistic accuracy $Acc^\star$, derived from CM*), and the area under the MCP (Multi–class Classification Performance) curve [28], [42]. The choice of the MCP curve is motivated by the fact that it is computed directly from the class-probability vectors returned by the classifier and is defined natively in the multi-class setting [43], rather than being obtained by reducing the problem to multiple binary tasks (e.g., one-vs-rest) and then applying micro/macro averaging, as is common in multi-class ROC analyses.

TABLE I
Description of Datasets From the UCI Repository: Dataset Name, Number of Samples (#s), Number of Variables (#v) and Number of Class Labels (#c), Respectively

Dataset	#s	#v	#c
australian credit	690	14	2
breast cancer coimbra	116	9	2
cardiotocography morphologic pattern	2,126	23	10
customer churn	3,150	13	2
data banknote authentication	1,372	4	2
divorce	170	54	2
fertility diagnosis	100	9	2
glass	214	9	6
heart	270	13	2
iris	150	4	3
landsat satellite	6,435	36	6
leaf	340	14	30
magic gamma telescope	19,020	10	2
movement libras	360	90	15
musk v1	476	166	2
musk v2	6,598	166	2
parkinson	195	22	2
phishing websites	11,055	30	2
planning relax	182	12	2
rice cammeo osmancik	3,810	7	2
room occupancy	10,129	16	4
sonar	208	60	2
speaker accent recognition	329	12	6
vertebral column	310	6	3
vowel	990	13	11
wine quality red	1,599	11	6
wine quality white	4,898	11	7
Mean	**2,788**	**31**	**5**

As a baseline classifier, the Random Forest (RF) algorithm [44] was chosen because it is an ensemble method rather than a simple model, typically achieves good results (highly competitive), is computationally efficient (built on fast decision trees), and can be considered a closed–box model. The performance of the RF classifier is evaluated using both traditional and probabilistic performance measures.

Table II reports the dataset-level results for RF. The mean hard-label accuracy is high ($Acc = 0.87$), indicating strong

TABLE II
RANDOM FOREST: DATASET, ACCURACY (Acc), CERTAINTY WEIGHT (λ_v), UNCERTAINTY WEIGHT (λ_u), PROBABILISTIC ACCURACY ($Acc^\star$), AND THE AU(MCP) CURVE, RESPECTIVELY

Dataset	Acc	λ_v	λ_u	$Acc^\star$	MCP
australian credit	0.87	0.84	0.16	0.79	0.70
breast cancer	0.72	0.72	0.28	0.63	0.56
cardiotocography	0.90	0.81	0.19	0.77	0.68
customer churn	0.96	0.94	0.06	0.93	0.88
data banknote	0.99	0.98	0.02	0.98	0.95
divorce	0.98	0.97	0.03	0.96	0.93
fertility diagnosis	0.86	0.88	0.12	0.80	0.73
glass	0.78	0.70	0.30	0.64	0.58
heart	0.83	0.79	0.21	0.73	0.64
iris	0.94	0.96	0.04	0.93	0.90
landsat satellite	0.92	0.86	0.14	0.84	0.78
leaf	0.81	0.58	0.42	0.54	0.49
magic gamma	0.88	0.85	0.15	0.81	0.73
movement libras	0.84	0.60	0.40	0.57	0.51
musk v1	0.90	0.78	0.22	0.75	0.65
musk v2	0.98	0.95	0.05	0.94	0.90
parkinson	0.91	0.87	0.13	0.84	0.76
phishing websites	0.97	0.96	0.04	0.95	0.91
planning relax	0.70	0.71	0.29	0.58	0.52
rice cammeo	0.92	0.93	0.07	0.89	0.85
room occupancy	1.00	0.99	0.01	0.99	0.98
sonar	0.82	0.72	0.28	0.69	0.61
speaker accent	0.81	0.63	0.37	0.59	0.54
vertebral column	0.84	0.82	0.18	0.77	0.70
vowel	0.97	0.70	0.30	0.70	0.61
winequality red	0.70	0.67	0.33	0.57	0.52
winequality white	0.70	0.65	0.35	0.56	0.51
Mean	**0.87**	**0.81**	**0.19**	**0.77**	**0.71**

TABLE III
MULTINOMIAL LOGISTIC REGRESSION WITH ISOTONIC CALIBRATION: DATASET, ACCURACY (Acc), CERTAINTY AND UNCERTAINTY WEIGHTS (λ_v AND λ_u), PROBABILISTIC ACCURACY ($Acc^\star$), AND THE AU(MCP) CURVE, RESPECTIVELY

Dataset	Acc	λ_v	λ_u	$Acc^\star$	MCP
australian credit	0.87	0.86	0.14	0.80	0.72
breast cancer	0.69	0.67	0.33	0.58	0.52
cardiotocography	0.70	0.57	0.43	0.50	0.46
customer churn	0.90	0.90	0.10	0.86	0.82
data banknote	0.99	0.99	0.01	0.99	0.99
divorce	0.98	0.97	0.03	0.96	0.93
fertility diagnosis	0.88	0.89	0.11	0.80	0.71
glass	0.60	0.56	0.44	0.47	0.44
heart	0.84	0.84	0.16	0.78	0.72
iris	0.95	0.82	0.18	0.80	0.72
landsat satellite	0.85	0.75	0.25	0.68	0.62
leaf	0.59	0.45	0.55	0.37	0.36
magic gamma	0.82	0.81	0.19	0.73	0.66
movement libras	0.63	0.43	0.57	0.36	0.36
musk v1	0.84	0.76	0.24	0.66	0.59
musk v2	0.91	0.86	0.14	0.82	0.72
parkinson	0.88	0.86	0.14	0.81	0.73
phishing websites	0.94	0.93	0.07	0.90	0.86
planning relax	0.71	0.71	0.29	0.59	0.52
rice cammeo	0.92	0.93	0.07	0.89	0.84
room occupancy	0.98	0.92	0.08	0.92	0.87
sonar	0.71	0.67	0.33	0.57	0.52
speaker accent	0.70	0.58	0.42	0.50	0.46
vertebral column	0.85	0.86	0.14	0.80	0.75
vowel	0.60	0.35	0.65	0.31	0.33
winequality red	0.59	0.53	0.47	0.44	0.42
winequality white	0.52	0.48	0.52	0.38	0.38
Mean	**0.79**	**0.74**	**0.26**	**0.68**	**0.63**

decision-level performance across the benchmark. However, the decisiveness diagnostic reveals that a non-negligible fraction of the predictive mass lies outside the top-1 component: on average, $\lambda_v = 0.81$ and $\lambda_u = 0.19$. This gap between accuracy and decisiveness is reflected in probability-aware summaries: $Acc^\star$ is typically below Acc, because $Acc^\star$ accounts for how much probability mass is assigned to the true class, not only whether the argmax is correct. This implies that despite high accuracy, RF may not always be reliable.

The dataset-wise ranges make this effect transparent. While Acc ranges from 0.70 to 1.00, $Acc^\star$ spans a wider range (0.54 to 0.99), indicating that similar decision accuracies can correspond to markedly different probability-mass support for the true class. For instance, on the *vowel* dataset, $Acc = 0.97$; however, because $\lambda_u = 0.30$ indicates substantial residual dispersion, the probabilistic accuracy drops to $Acc^\star = 0.70$. Conversely, in highly decisive settings such as *room occupancy*, Acc and $Acc^\star$ nearly coincide (1.00 vs. 0.99) and λ_u is minimal (0.01).

Finally, MCP ranges from 0.49 to 0.98 and exhibits greater between-dataset variability than Acc. In high-uncertainty cases MCP departs markedly from Acc (e.g., *vowel*, MCP$= 0.61$), whereas it remains close to decision accuracy when predictions are highly decisive (e.g., *room occupancy*, MCP$= 0.98$). Overall, MCP typically lies closer to $Acc^\star$ than to Acc, which is consistent with both being probability-aware summaries.

To support the classifier-agnostic character of the framework, Table III reports the same indicators when Q is obtained from multinomial logistic regression (MLR) with isotonic post-hoc calibration. Isotonic calibration is fitted using only training-fold data (with an inner 3-fold cross-validation), and the reported test-fold probabilities are obtained from the calibrated model.

The main qualitative behaviours persist: probability-aware measures are more conservative than hard-label accuracy whenever predictions are not highly decisive. On average, MLR attains $Acc \approx 0.79$ with a lower decisiveness weight ($\lambda_v \approx 0.74$) and a larger residual mass ($\lambda_u \approx 0.26$), leading to smaller probability-aware values ($Acc^\star \approx 0.68$, MCP≈ 0.63).

As with RF, heterogeneity across datasets is substantial. Accuracy ranges from 0.52 to 0.99, whereas $Acc^\star$ ranges from 0.31 to 0.99, indicating that probability-mass support can deteriorate substantially in difficult multi-class problems even when

TABLE IV
COMPARISON OF RANDOM FOREST (RF) AND MULTINOMIAL LOGISTIC REGRESSION (MLR), WITH AND WITHOUT ISOTONIC CALIBRATION. VALUES ARE THE MEAN ± STANDARD DEVIATION ACROSS THE 27 DATASETS; THE 95% CONFIDENCE INTERVAL FOR THE MEAN IS SHOWN IN BRACKETS.

Classifier	Calibration	Acc	$Acc^\star$	λ_v	MCP
RF	None	0.870 ± 0.094 [0.832, 0.907]	0.768 ± 0.147 [0.709, 0.826]	0.811 ± 0.128 [0.760, 0.861]	0.708 ± 0.158 [0.646, 0.771]
RF	Isotonic	0.866 ± 0.092 [0.830, 0.902]	0.775 ± 0.144 [0.718, 0.832]	0.825 ± 0.121 [0.777, 0.873]	0.713 ± 0.152 [0.653, 0.774]
MLR	None	0.823 ± 0.119 [0.775, 0.870]	0.740 ± 0.169 [0.673, 0.806]	0.809 ± 0.141 [0.753, 0.864]	0.686 ± 0.163 [0.621, 0.751]
MLR	Isotonic	0.793 ± 0.142 [0.737, 0.849]	0.677 ± 0.203 [0.597, 0.758]	0.737 ± 0.184 [0.664, 0.810]	0.628 ± 0.189 [0.554, 0.703]

decision accuracy is moderate. This is most evident in highly diffuse cases such as *vowel* ($\lambda_u = 0.65$), where both $Acc^\star$ and MCP drop sharply relative to Acc. In contrast, near-deterministic behaviour is again observed in *data banknote*, where $\lambda_u \approx 0.01$ and probability-aware measures approach decision accuracy.

Across heterogeneous datasets and two distinct probability generators, the probabilistic confusion matrix $\mathrm{CM}^\star$ reveals information that is suppressed by the hard projection underlying CM, namely the amount of probability mass supporting the true class and the overall diffuseness of the predictive distributions. Accordingly, when λ_u is non-negligible, probability-aware summaries ($Acc^\star$, MCP) can be markedly lower than Acc even if argmax decisions are frequently correct.

The weaker results of MLR relative to RF are most naturally attributed to differences in hypothesis class and inductive bias, rather than to any limitation of the probabilistic framework. Multinomial logistic regression implements linear decision boundaries in the original feature space: for any pair of classes, the decision surface is a hyperplane. Random Forests, by contrast, realise highly non-linear decision functions via ensembles of axis-aligned partitions and can capture feature interactions without explicit feature engineering. Consequently, on heterogeneous datasets where class separation is not approximately linear (particularly in multi-class settings with overlapping classes or interaction-driven structure), MLR may underfit and yield both lower decision accuracy and more diffuse predictive distributions (lower λ_v, higher λ_u). Post-hoc calibration (e.g., isotonic regression) can improve probability-outcome alignment but does not increase discriminative capacity, and therefore cannot in general close performance gaps that arise from representational constraints.

Table IV provides a compact comparison of decision-level performance, probability-mass performance, and predictive decisiveness across the two classifier families and calibration conditions. In all four cases, $Acc^\star$ and MCP remain below hard-label accuracy, confirming that correct argmax decisions are not necessarily supported by equally strong probability mass.

For RF, isotonic calibration leaves hard-label accuracy essentially unchanged, while producing small increases in λ_v, $Acc^\star$, and MCP. Thus, the calibrated RF probabilities are slightly more concentrated and allocate marginally more mass to the true classes, although the substantial overlap between confidence intervals indicates that these differences should be interpreted descriptively.

The effect is different for MLR. Isotonic calibration reduces Acc, λ_v, $Acc^\star$, and MCP, indicating less decisive predictive distributions and weaker probability-mass support for the true classes under the present experimental protocol. This result also shows that post-hoc calibration does not necessarily improve every probability-aware performance quantity.

Notably, the uncalibrated RF and MLR models have almost identical mean decisiveness weights (0.811 and 0.809, respectively), while RF achieves higher Acc, $Acc^\star$, and MCP. This confirms that λ_v measures predictive concentration rather than correctness: models may be similarly decisive while differing substantially in both classification performance and probability-mass support for the true classes.

Overall, these results reinforce the central message of the paper: $\mathrm{CM}^\star$ and the $\mathrm{V/U}$ decomposition provide a principled, confusion-matrix-structured account of decisiveness that is not captured by hard-label evaluation and remains informative across classifier families.

The relationship between the certainty weight (λ_v) and the gap between nominal and probabilistic performance was analysed by correlating λ_v with the difference $Acc - \mathrm{MCP}$ (see Fig. 5). Spearman's ρ is a non-parametric statistic that evaluates the strength of a monotonic relationship between ranked values, without assuming linearity or specific distributional properties of the data. This makes it particularly suitable for assessing general dependencies in heterogeneous datasets. The Spearman rank correlation remained strongly negative ($\rho = -0.91, -0.87, -0.86$, and -0.77 for raw RF, isotonic RF, raw MLR, and isotonic MLR, respectively; all $p < 3 \times 10^{-6}$). This indicates that, across the considered datasets, larger decisiveness (higher λ_v) tends to co-occur with smaller discrepancies between hard-label accuracy and a probability-aware summary. This is a descriptive association and should not be interpreted causally. Moreover, both λ_v and $Acc - \mathrm{MCP}$ may be influenced by shared dataset-level factors—for example, the number of classes k, class imbalance, sample size, and intrinsic dataset separability—which can affect both the concentration of predicted probabilities and the relationship between hard-label and

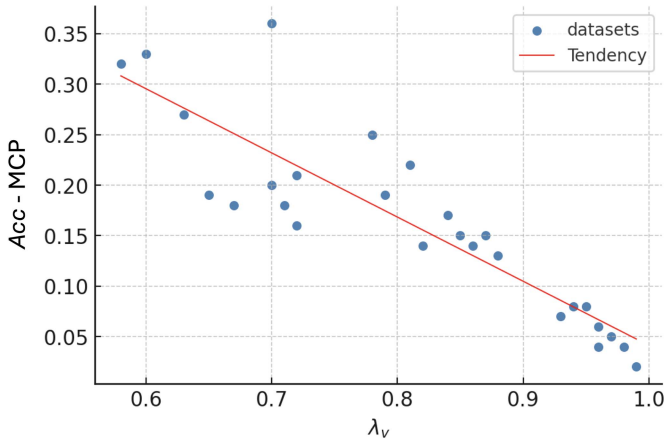


Fig. 5. Correlation between certainty weight and the difference between accuracy and the area under the MCP curve ($Acc - \text{MCP}$) for Random Forest without calibration.

probability-aware performance summaries. Accordingly, Fig. 5 is best read as a compact empirical regularity supporting the intuition that when predictions are more decisive, hard-label and probability-aware summaries tend to be closer, whereas more diffuse predictions are associated with larger gaps.

VII. Conclusion

This work introduces a probabilistic, reliability–aware evaluation framework for classification that complements the hard–label confusion matrix (CM) with its probabilistic counterpart (CM*) and decomposes CM* into decisive (V) and residual-dispersion (U) components. The framework is model-agnostic and measure-agnostic: any confusion-matrix-based metric can be computed on CM* without changing its definition, thereby retaining familiar performance criteria while accounting for predictive probability mass rather than hard binarised decisions.

Empirically, on 27 UCI datasets, probability-aware summaries are systematically more conservative than hard-label accuracy. For the Random Forest baseline, the mean Acc is 0.87, whereas $Acc^\star$ and MCP average 0.77 and 0.71, respectively, with a mean certainty weight $\lambda_v = 0.81$ (and $\lambda_u = 0.19$). These results indicate that a non-trivial fraction of predictive mass lies outside the decisive top-1 component, so decision-level accuracy can overstate reliability. Moreover, the diagnostic role of λ_v is consistent across datasets: higher decisiveness is associated with smaller disagreement between decision-level and probability-aware summaries (e.g., between Acc and MCP). The Spearman rank correlation between λ_v and $Acc-\text{MCP}$ is -0.91, evidencing a strong monotonic relationship. These findings show that models with higher certainty deliver scores whose nominal accuracy better reflects their probabilistic reliability.

To support classifier-agnosticity in practice, we additionally reported results using a second probability generator, multinomial logistic regression with isotonic calibration. The same qualitative behaviours persist: when λ_u is not small, probability-aware measures ($Acc^\star$, MCP) can be substantially lower than Acc even if argmax decisions are often correct, highlighting reduced decisiveness and weaker probability-mass support for the true class. This shows that the information revealed by CM* and by the V/U decomposition is not specific to a particular classifier family.

Since CM* is constructed directly from Q, its numerical values necessarily depend on the quality of the predicted probabilities. If Q is miscalibrated, overconfident, or otherwise unreliable, CM* will faithfully summarise that probability allocation, but CM*-based measures cannot by themselves diagnose the calibration defect. In problems with many classes, there is no intrinsic instability; however, top-1 probabilities often decrease as k increases, so decisiveness should be interpreted relative to the uniform baseline $1/k$.

Our framework retains the familiar confusion-matrix perspective while extending it to probability-mass outputs: the probabilistic confusion matrix CM* preserves the full class-probability mass and reduces to the standard confusion matrix CM under argmax binarisation. Thus, CM* complements CM by augmenting decision-level reporting with a structured account of predictive probability mass, which is particularly informative when decisiveness, residual dispersion, or probabilistic reliability are relevant.

We therefore recommend computing standard metrics on CM* and reporting the associated decisiveness and residual-dispersion summaries alongside conventional CM-based measures when class-probability outputs are available.

Future work will examine additional confusion-matrix-derived measures under the proposed decomposition in the multi-class regime and pursue a tighter integration with explainability and probability-quality diagnostics for critical applications.

Source Code

The software implementation and illustrative examples are publicly available at https://github.com/TIC200LAB/Quantifying-Predictive-Uncertainty.

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